# Internship Project Brief: Forecasting Demand for National Research Supercomputing

Master of Business Analytics internship · 12 weeks · Australian Academy of Science

## Purpose

Develop a robust forecast of Australian research demand for high-performance computing (HPC) to 2030, to inform national policy discussions for national public compute investment. The project combines data engineering, descriptive analytics, and structural forecasting on real national datasets, with a policy-facing deliverable.

## Proposed project phases

- **Weeks 1–3 — Data assembly.** Build a longitudinal dataset from NCMAS outcomes (multi-year CSVs), requested-vs-awarded figures where published, and NCI/Pawsey annual reports (utilisation, capacity). Map the full allocation landscape (merit schemes, partner shares, Tier 2 (institutional) and Tier 1 (national) allocations). Classify projects by field of research. Devise and document a normalisation strategy for service units across facilities and hardware generations, including CPU vs GPU hours.
- **Weeks 3–6 — Descriptive analysis.** Oversubscription trends; allocation concentration (test whether minimum-award floors indicate rationing of compute resources); field-of-research mix over time; institutional distribution; and the pace and composition of the CPU-to-GPU shift as a leading indicator of AI-driven demand. Mid-point checkpoint presentation before modelling begins. Expert review should be conducted at suitable stages to gather feedback on indicators, proposed model and outputs.
- **Weeks 6–10 — Demand modelling.** Driver-based structural model rather than pure time-series methods: research workforce size, competitive grant flows in compute-intensive fields (as a 1–2 year leading indicator), and per-project compute-intensity growth by field. Benchmark against uncensored international comparators (e.g. EuroHPC, NSF ACCESS, UK ARCHER2) on compute per researcher and per research dollar.
- **Weeks 10–12 — Scenarios and reporting.** Minimum three scenarios (status quo; restored national R&D intensity; AI acceleration), each expressed as a capacity gap in facility-equivalent terms. Final write-up and handover.

## The analytical challenge

Australia's merit allocation schemes (e.g. National Computational Merit Allocation Scheme) publish what was *awarded*, not what was *needed*. Allocations are supply-constrained outputs of a competitive process: schemes are oversubscribed, researchers self-limit requests to fundable amounts, and some never apply at all. Naive extrapolation of allocation time series therefore forecasts supply, not demand. The project task is to estimate latent demand via inclusion of different factors/indicators (e.g. oversubscription ratios, funding-flow leading indicators, and international benchmarking of comparators).

## Deliverables

- Cleaned, documented longitudinal dataset (a lasting asset independent of the forecast)
- Technical report: methodology, model, assumptions, sensitivity analysis, limitations
- Two-page policy summary translating the forecast into advocacy-usable figures
- Mid-point (descriptive findings) and final presentations

## Key data sources

- NCMAS outcomes and scheme reports (my.nci.org.au)
- NCI and Pawsey annual reports
- ARC / NHMRC / MRFF grant outcome data; Department of Education research workforce statistics
- International: EuroHPC, NSF ACCESS, ARCHER2 allocation and utilisation reports

## Methodological cautions

- Service units are not comparable across facilities or years without explicit normalisation; GPU and CPU hours are not interchangeable.
- Capacity events (e.g. new systems coming online) create step changes in allocations that are not demand growth and must be modelled as supply shocks.
- The observed series is short and censored; model choice should favour interpretable structural drivers over black-box time-series fitting, with uncertainty and assumptions stated clearly.

## Background

The Academy hosted an expert roundtable on high-performance computing and data (HPCD) in 2023.

Following the roundtable, the Academy published:

- The future computing needs of the Australian science sector (2024) – a brief based on input received during the roundtable. It includes an overview of Australia's current HPCD landscape and future demand.
- Bringing Australia's supercomputing up to speed (2025) – a proposal calling for the Australian Government to invest in a national strategy for next-generation HPCD.

In 2025, the Academy released a series of papers on artificial intelligence and science. The infrastructure paper discusses HPCD for AI.

### Australia's supercomputing national research infrastructure

Australia hosts two national (Tier-1) HPCD facilities that are accessible to researchers: NCI Australia and Pawsey Supercomputing Research Centre. The supercomputing brief linked above provides some background on how NCI and Pawsey fit into the current landscape of other supercomputing infrastructure. Australia also has several Tier 2 HPCD facilities based at different research institutions.

### Current policy context

This work will inform the Academy's input on the National Research Infrastructure Roadmap, and advice to government on development of Australian Standards for AI and implementing the Data Centre Expectations.

### Science capability demand forecasting

The Academy's Australian Science, Australia's Future initiative analysed Australia's science capability to meet national challenges.

One stream of evidence that helped us assess science capability was analysis by Academy Fellow Professor Rob Hyndman and a previous MBAT student Kelly Ngyuen to forecast the age structure of science disciplines.

A journal article on the age structure forecasts was published this year and the code is available on GitHub.

See Appendix A of the Australian Science, Australia's Future report for the full methodology and how the forecasts were used alongside other streams of evidence for our analysis.